Case 3: Time-independent efficiency and production cost

!! Employees with different levels of productivity produce different quantities of goods/services in a given amount of time.

At Company X, where products are manufactured manually, **workers are paid 2 $** for each unit they produce.

It is observed that when a worker produces 80 units per day, the manufacturing defect rate is 5%.
This manufacturing defect rate rises to 10% when 90 units are produced per day, and to 20% when 100 units are produced per day.

Workers are not paid for defective pieces.

The material cost per unit is 2.5 $, and the materials from defective pieces cannot be reused.

The fixed cost per worker is 50 $/day.

When calculating labor costs, the weekly working time is assumed to be 45 hours, and a month is considered to consist of 4 weeks. Given this:

a) At which of the three different production volumes do workers receive the highest total payment?
b) At which production volume does the company operate with the lowest unit cost?
c) Is it necessary to make adjustments to the wages?

Answer:

a) Wages received by workers at three different production levels;

Daily production (units)	80	**90**	100
Defective pieces	4	**9**	20
Accepted pieces	76	**81**	80
Wages	2 x 76 = 152 $	**2 x 81= 162 $**	2 x 80 = 160 $

b) the production volume at which the company operates with the lowest unit cost;

Daily production (units)	**80**	90	100
Accepted pieces	**76**	81	80
Material Cost	**2.5 x 80 = 200 $**	2.5 x 90 = 225$	2.5 x 100=250$
Labour Cost	**2 x 76 = 152 $**	2 x 81 = 162$	2 x 80 = 160$

Fixed Cost	**50$**	50$	50$
TOTAL COST	**200+152+50=402$**	225+162+50=437$	250+160+50=460$
Cost per accepted units	**402 / 76 =5.29 $**	437 / 81= 5.40 $	460 / 80 =5.75$

c) As can be seen, the interests of the employer and the workers do not align; while it is in the workers' interest to produce 90 units per day, the employer achieves minimum production costs when 80 units are produced daily.